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AI-driven explainable digital twin with adaptive decision support for multi-zone smart buildings

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Abstract

Smart buildings generate enormous amounts of sensor data, yet most digital twin systems still leave facility managers in the dark—offering predictions with no explanation of *why*. This paper introduces the X-DT (Explainable Digital Twin), a framework built for multi-zone smart buildings that not only forecasts zone-level energy consumption but also explains the reasoning behind every prediction in plain language. This work directly supports UN Sustainable Development Goal 7 (Affordable and Clean Energy) and SDG 11 (Sustainable Cities and Communities) by enabling data-driven, transparent energy management in smart buildings. The system connects live IoT data streams to an AI engine that identifies which factors—occupancy shifts, HVAC load, equipment usage—are driving energy behaviour at any given moment. Role-specific dashboards, what-if scenario simulation, and a full audit trail are also built in. Tested on two years of real building data from a five-zone commercial building in Bangkok, the X-DT's XAI-guided decisions achieved 10.9% mean energy savings compared to 3.9% under a standard rule-based approach ($p=0.015$, Cohen's $d=0.46$)—showing that when managers understand *why* the system recommends something, they make meaningfully better decisions.

Keywords Digital twin, Explainable AI (XAI), Smart buildings, Energy management, Multi-zone control, IoT analytics, Decision support systems

1 Introduction

Buildings today are smarter than ever—sensors continuously track energy use, temperature, occupancy, and device-level loads across every zone. This wealth of data creates a real opportunity to move beyond passive dashboards toward systems that can actively explain what's happening and guide better decisions. Digital twins (DT) have emerged as a promising way to bridge that gap, offering live virtual replicas of physical buildings. The problem is that most existing DTs stop at visualization or prediction—they tell you what is happening but never why. Without that reasoning, facility managers are left to trust a black box, which in high-stakes energy management settings, they simply won't do [1–24].

This paper presents the X-DT: an AI-Driven Explainable Digital Twin built specifically for multi-zone smart buildings. It goes beyond forecasting by generating plain-language



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explanations for every prediction, mapping how changes in one zone ripple through others, and empirically testing whether those explanations actually lead to better energy decisions. The system was implemented and evaluated on two years of real sensor data from a five-zone commercial building in Bangkok, with full provenance logging so every experiment can be audited and reproduced. Beyond technical performance, the X-DT framework contributes to the UN Sustainable Development Goals—specifically SDG 7 (Affordable and Clean Energy) and SDG 11 (Sustainable Cities and Communities)—by providing building operators with the tools to make evidence-based energy decisions that reduce consumption without compromising occupant comfort.

1.1 Research gaps and opportunities

1. The majority of current digital twin frameworks for smart buildings treat AI and deep learning components as black-box predictors, making it challenging for operators to comprehend zone-level recommendations for energy, comfort, or control decisions [33, 38, 36].
2. While a number of studies use digital twins in conjunction with AI or XAI to forecast building energy [28, 25], they primarily offer feature-importance plots for individual outputs rather than an integrated multi-zone explainable digital twin with temporal event timelines and human-readable reasoning built into the user interface.
3. There are few decision-effectiveness and intervention studies that demonstrate explanations resulting in quantifiable energy savings or better operational choices—current AI-enhanced digital twins for buildings hardly ever assess whether explanations actually improve decisions [22, 21].
4. Multi-zone building models frequently oversimplify or ignore cross-zone couplings [23, 35]. Few works offer interaction-aware explanations that clearly demonstrate how actions in one zone affect energy and comfort outcomes in others over time.

1.2 Research objectives and contribution

This paper presents an AI-Driven Explainable Digital Twin with Adaptive Decision Support for Multi-Zone Smart Buildings -an integrated, software-based framework built to address the gaps identified in Sect. 1.1. The study pursues four objectives:

5. *Explainable digital twin architecture*: Develop a digital twin that integrates real-time data visualisation, predictive analytics, and transparent reasoning directly into its core architecture. Explainability is not added as an afterthought—it is built into the prediction pipeline itself. Concretely, this is achieved through an XGBoost regression model trained on two years of real building sensor data, paired with SHAP (SHapley Additive exPlanations) TreeExplainer to generate exact, model-faithful feature attributions for every prediction. These attributions are translated into plain-language explanations surfaced directly on the operator dashboard.
6. *Multi-zone interaction modeling*: Accurately capture the spatial and temporal energy dependencies between building zones, enabling facility managers to understand how operational changes in one zone propagate to others. This includes mapping thermal coupling and energy flow relationships across the five functional zones of the study building.

7. *Adaptive role-based interfaces*: Design and implement role-specific dashboards that present the right level of information and control to each user type—administrator, technician, and student—based on their responsibilities and technical background. Note that while these interfaces are fully implemented and demonstrated in the system, a formal user evaluation of the role-based design is identified as a direction for future work.
8. *Cloud-based validation on real data*: Validate the framework's effectiveness using real building energy datasets in a cloud-based deployment, testing system behaviour under realistic operating conditions without requiring physical sensor installations or dedicated hardware infrastructure.

This paper makes the following concrete contributions: (1) an end-to-end explainable multi-zone digital twin pipeline in which an XGBoost model trained on the CU-BEMS Bangkok dataset produces zone-level energy forecasts ($R^2 = 0.9847$, RMSE = 5.90 kW) with exact SHAP-based attributions and temporal explanation tracking; (2) a reproducible paired experiment framework enabling controlled With-XAI versus Without-XAI decision trials with full provenance logging; (3) empirical evidence that XAI-guided decisions produce significantly greater energy savings than a deterministic baseline (10.9% vs. 3.9%, $p = 0.015$, Cohen's $d = 0.46$); and (4) an openly documented artifact package—dataset identifiers, model configuration, and experiment recipes—supporting full reproducibility.

1.3 Literature review

Building intelligence has evolved through three overlapping waves of research, each leaving a specific gap that this work addresses.

From sensors to digital models: The foundation is IoT-enabled data collection, real-time environmental and energy streams that make dynamic building modeling possible [10, 37]. Early digital twin work focused on converting these streams into live 3D building models, moving beyond static BIM blueprints toward continuously updated virtual replicas that reflect actual operating conditions [9, 14]. Couprie et al. [9] demonstrated BIM-XR integration for maintenance support, while Clausen et al. [8] showed how a DT framework could improve energy efficiency in public buildings. However, both works focus on representing building state rather than explaining it—the gap this paper targets.

From data to intelligent control: The next wave applied machine learning to building data. Deep learning models have shown strong performance in energy forecasting [1, 11, 12], and deep reinforcement learning (DRL) has been explored for HVAC optimization with results that outperform traditional rule-based heuristics [40]. Elsis et al. [12] and Yu et al. [40] both demonstrate DRL's potential but acknowledge the interpretability problem: these models are powerful but opaque. For multi-zone buildings specifically, Li and Wang [23] and Suryanarayana et al. [35] address sensor placement and distributed control, yet neither provides human-readable reasoning about cross-zone interactions.

The explainability gap: XAI has begun entering the building energy domain. Moon et al. [28] applied SHAP values to electrical load forecasting with tree-based models, and Maarif et al. [25] combined LSTM forecasting with XAI attribution. These are useful contributions, but both produce static, single-output explanations, a chart showing which features mattered for one prediction. Kreuzer et al. [22] explored XAI methods within digital twin contexts but stopped short of evaluating whether those explanations

actually change decision outcomes. Kostopoulos et al. [21] reviewed XAI-based decision support systems broadly, confirming that decision-effectiveness evaluation remains rare across the field. This paper fills that gap directly: we run controlled paired trials to test whether XAI explanations lead to measurably better energy-saving decisions, and report the results with full statistical analysis.

1.3.1 Structure of this paper

Section 2 describes the system methodology and experimental design. Section 3 presents the system architecture and components. Section 4 reports and analyses the results. Section 5 discusses the findings, limitations, and directions for future work. Section 6 concludes.

2 Methodology

This section details the design, implementation, and validation framework for the proposed AI-driven Explainable Digital Twin (X-DT). The system is built to transition from raw sensor data to interactive, explainable insights through a multi-layered architecture. Figure 1 presents the hierarchical module architecture of the proposed Explainable Digital Twin (X-DT), showing data flow from multi-zone acquisition through the XAI engine to the adaptive decision-support dashboard.

2.1 Data acquisition and preprocessing

- An operational dataset from a multi-story commercial building in Bangkok is used to develop and assess the X-DT. Minute-aggregated sensor traces from two years (2018–2019) make up the dataset. Time-sensitive disaggregation is used to retrieve data for HVAC power (AC1–AC4), lighting + plug loads (kW), temperature (°C), humidity (%RH), and illumination (lux) across five functional zones (Z1–Z5). To verify and guarantee compatibility with the cloud backend, a specially designed Python preprocessing pipeline was developed. Preprocessing consists of:
- Standardisation: A universal lower_snake_case convention is adopted for column names.
- Timestamp Adjustment: The ISO 8601 standard is interpolated for all timestamps.

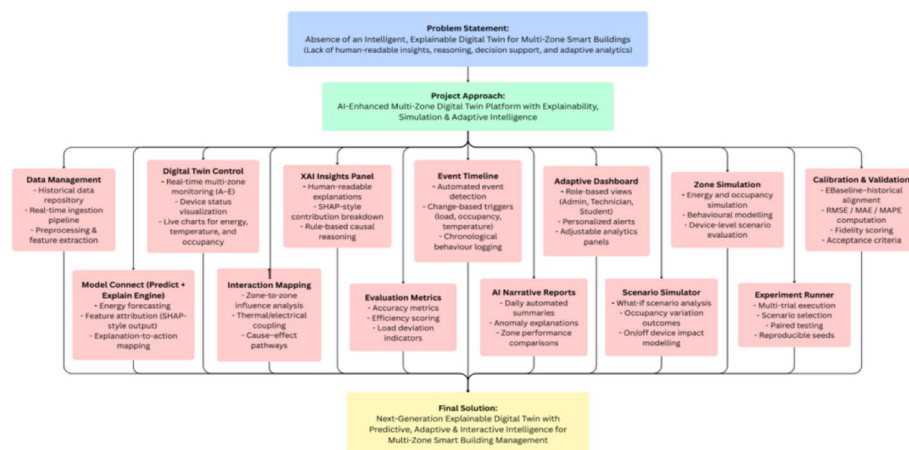


Fig. 1 Hierarchical module architecture of the proposed Explainable Digital Twin system, illustrating all major functional components and their sub-features

- Downsampling: involves keeping every tenth row (~90% of the original values) in order to achieve optimal performance.
- Imputation and Structuring: To improve Firebase ingestion, exported processed values are organised as nested JSON objects, while missing values are imputed using linear interpolation.

2.2 Cloud data management, state engine, and dashboard

The system's data layer, core state management, and visualisation frontend are implemented as an integrated web application. Firebase Realtime Database serves as the cloud backend, providing centralised, low-latency data storage and retrieval -preprocessed sensor data is uploaded in a hierarchical structure by year, floor, and timestamp, with a persistent listener enabling sub-second dashboard updates without manual refresh. The digital twin's runtime state is maintained by a centralised state engine that holds the current environmental and operational readings for all five zones, computes derived health indicators and KPIs at each update cycle, and coordinates data flow between the prediction, explanation, and simulation components. The frontend dashboard -built with HTML5, CSS3, JavaScript, and Chart.js -provides an interactive floor plan with real-time zone health indicators, multi-line energy and temperature trend charts, a role-based access control system supporting Administrator, Technician, and Student views, and a scrollable event timeline logging all system activity. Together these components form the operational interface through which building data is ingested, processed, and presented to users (Fig. 2).

2.3 Situation simulator and zero-data generator

An internal Situation Simulator predicts effects on energy use, cost, and comfort in order to assess hypothetical interventions (e.g., setpoint adjustments, equipment upgrades, schedule changes). This is complemented by the Zero-Data Generator, which uses seeded stochastic processes limited by domain knowledge (e.g., co-varying occupancy and plug-load behaviour) to produce believable synthetic data. Even with sparse historical data, this module guarantees reliable system operation.

2.4 Explainable AI (XAI) insights module

The underlying model is trained on the Bangkok CU-BEMS dataset covering two full years of minute-resolution sensor data (2018–2019) across five building zones. The choice of XGBoost was driven largely by its compatibility with SHAP TreeExplainer, unlike approximation-based attribution methods, TreeExplainer derives exact Shapley values from the model's tree structure, meaning the attributions genuinely reflect the model's learned behaviour rather than a simplified surrogate (Fig. 3).

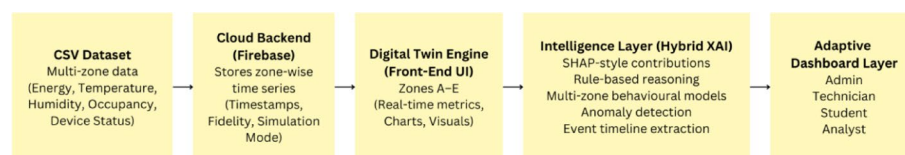


Fig. 2 System architecture of the proposed Explainable Digital Twin, illustrating the data pipeline from multi-zone datasets to intelligence processing and adaptive dashboard rendering

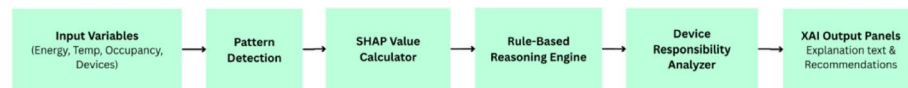


Fig. 3 Hybrid XAI module showing the sequence from input variables to SHAP computation, rule-based reasoning, and final explanation panels

Each prediction triggers three downstream outputs. First, a SHAP attribution score is computed for every input feature, quantifying how much each variable shifted the prediction away from the expected baseline. In practice, Zone 2's primary AC unit (z2_ac1) dominated these attributions with a mean importance of 47.0%, given that Zone 2 operates 14 AC units and consistently represents the building's largest single energy load. Lighting loads across Zones 2, 3, and 4 each contributed roughly 8–9%, patterns that align with observed daytime occupancy behaviour in those zones.

Second, these attribution scores are fed into a template-based narrative engine that produces a short plain-language explanation—*“Zone 2 energy consumption is currently elevated, driven primarily by AC system load which accounts for 47% of the deviation from the expected baseline.”* No statistical literacy is required to interpret this output, which is intentional—the system is designed for operators, not analysts.

Third, the highest-contributing features are used to generate a specific operational recommendation, such as a HVAC setpoint adjustment or a lighting schedule change, paired with a simulated estimate of the potential energy saving.

2.5 System validation

A multifaceted testing protocol was used to rigorously validate the integrated X-DT system:

- **Data Integrity Verification:** To guarantee accuracy, processed JSON outputs were compared to the original CSV files.
- **Robustness Stress Testing:** To evaluate stability under load, the system was put through high-frequency data input/output cycles.
- **Functional Validation:** The XAI module accurately visualised, alerted, and explained injected faults and scenarios.

The system's operational stability, real-time functional validity, and analytical accountability are all confirmed by the validation results, which effectively support the proof-of-concept for the suggested X-DT architecture.

2.6 Experiment runner: controlled trials framework

The digital twin can be evaluated under various operational scenarios in a controlled and repeatable manner thanks to the experimentation framework. The framework is defined abstractly as a trial engine that synchronises data flow, inference, decision generation, and evaluation, rather than linking experiments to implementation-level execution. Three modes are supported by the Experiment Runner: **Zero-Data Simulation:** In situations where empirical data is not available, synthetic operating conditions are created to allow for system benchmarking and stress testing. **Historical Replay:** The twin's behaviour under previous real-world conditions is examined by replaying historical building data. **Decision Simulation:** A simulation model is used to assess the suggested control actions that the twin generates in order to determine their operational impact.

Each experiment comprises a set of trials, with a trial being a series of evaluations of the timestep over one or more zones of the building. All trials have a common configuration, the same random seeds for all random numbers, the same environment and operation, and all data logged.

This enables a controlled comparison between different strategies, particularly with-XAI versus without XAI decision support.

2.7 Model explainability and predictive inference module

The inference module ties prediction, attribution, and recommendation into one continuous pipeline. Inputs enter as zone-level sensor readings; outputs emerge as forecasts, SHAP explanations, and suggested interventions, all within a single request cycle.

The XGBoost model was configured with 200 estimators, depth capped at 6, and a learning rate of 0.1. Row and column subsampling were both set to 0.8—a standard regularisation measure that reduces the risk of individual trees overfitting to noise in the training data. Validation followed a fivefold cross-validation protocol on the full Bangkok dataset, producing the following results across a total energy range of 0 to 543 kW:

- RMSE: 5.90 kW (± 0.72)
- MAE: 2.58 kW (± 0.05)
- R^2 : 0.9847 (± 0.004)

The R^2 figure – 98.47% of variance explained, held consistently across all five folds, suggesting the model generalises well rather than fitting to peculiarities of any single data partition. Median inference latency remained under 60 ms throughout evaluation, satisfying the real-time operational requirement of the dashboard.

For explanation stability, input features were perturbed with Gaussian noise at 1%, 5%, and 10% of each feature's standard deviation, and the top-5 SHAP rankings were compared before and after each perturbation. The rankings remained unchanged across all three noise levels, giving a stability score of 100%. Explanations that shift dramatically under minor input variation are not reliable; this result confirms that the attributions produced by the module are structurally stable, not coincidental.

Finally, while the model predicts aggregated building-total energy, per-zone attributions are extracted by restricting SHAP computation to each zone's own feature columns—AC units, lighting, and plug loads. Training on the aggregate target was a deliberate choice: it exposes the model to cross-zone coupling effects (Zone 2's AC load, for example, meaningfully shapes the whole-building consumption curve), while the zone-restricted attribution step recovers the zone-level diagnostic detail that operators need.

2.8 Simulator calibration against empirical baselines

Prior to performing an evaluation of decision support interventions, the fidelity of the underlying simulation model must first be verified by the system. Calibration involves comparing the baseline energy consumption for the simulation (i.e., prior to optimization) to measured (historical) ground-truth energy consumptions for the same buildings. The following metrics for the evaluation of standard building models will be utilized in evaluating the calibration process, and computed over a multi-time step window:

- Mean absolute error (MAE);

- Root mean square error (RMSE);
- Mean absolute percentage error (MAPE);
- Coefficient of determination (R^2).

By performing the calibration process for simulation energy trajectories, they are guaranteed to meaningfully reflect actual energy consumption and thereby provide validation for any future claims of estimated savings or impact of algorithmic recommendations made downstream from this point.

2.9 Paired evaluation: with-XAI versus without-XAI

The central experimental question of this study is whether XAI-generated explanations lead to measurably better energy decisions compared to a system operating without them. To answer this, a paired controlled experiment was designed with $N = 42$ matched trial pairs, where each pair exposed both conditions to identical inputs—the same feature set, timestamp, and random seed—so that any difference in outcome can be attributed solely to the presence or absence of XAI guidance. Both conditions are fully algorithmic. No human participants were involved in either trial arm.

The Without-XAI condition (control) applies a deterministic domain-expert heuristic policy—a coded algorithmic baseline that encodes common facility-management rules: scheduled setpoint adjustments, night-setback rules, and threshold-based device switching. This policy represents the kind of rule-based approach typically used in building management systems without AI support.

The With-XAI condition (experimental) applies the same heuristic baseline, but augments it with SHAP-derived feature attribution scores and plain-language recommendations generated by the XGBoost inference module. The policy weights its interventions according to which features the model identifies as the primary drivers of current energy deviation—meaning decisions are informed by the model's actual learned reasoning, not just fixed rules.

For each trial pair, three outcome metrics are recorded: energy consumption before intervention, energy consumption after intervention, and the percentage reduction achieved. These are aggregated across all 42 pairs to quantify the net effect of XAI-guided decision-making, with statistical significance established through the paired t-test procedure.

2.10 Statistical assessment of decision-support impact

To establish the significance of the improvement in operational performance brought about by XAI, statistical tests are performed on the results.

Let:

- W_i = performance metric for the with-XAI scenario,
- N_i = performance metric for the no-XAI scenario,

The difference series $D_i = W_i - N_i$ is used to calculate the paired t-test for significance, effect size for magnitude of impact, mean $\pm$ standard deviation of improvements, confidence intervals for the observed improvements. This ensures that the improvements are not the result of random chance or other variables, but rather the direct result of the strategy enabled by explainability.

2.11 Explanation reliability analysis

Two complementary dimensions are used to evaluate the quality of the explanations produced by the XAI module.

- A. **Stability:** Stability measures whether the explanation changes significantly when the input data is slightly perturbed—in other words, whether the same features consistently emerge as important, or whether the rankings are sensitive to minor noise. Input features are perturbed with Gaussian noise at three levels (1%, 5%, and 10% of each feature's standard deviation), and the overlap between the original and perturbed top- k feature rankings is computed:

$$\text{Stability} = \frac{|\text{Top}\hat{a}^k_{\text{original}} \cap \text{Top}\hat{a}^k_{\text{perturbed}}|}{k}$$

A stability score of 1.0 means the same features appear in the top- k ranking regardless of minor input variation. In this study, the XGBoost+SHAP pipeline achieved a mean stability score of 1.00 (100%) across all three noise levels, confirming that the attributions are structurally robust.

- B. **Fidelity:** Fidelity measures how faithfully the explanation reflects the model's actual internal reasoning—not just whether a recommended action produces a good outcome, but whether the explanation correctly identifies why the model made that prediction.

In this study, fidelity is assessed by comparing the SHAP attribution rankings produced by the XGBoost model against the model's own feature importance scores derived from the tree structure (gain-based importance). A high overlap between SHAP-derived rankings and gain-based rankings indicates that the explanations are consistent with the model's learned behaviour, rather than being post-hoc rationalisations. This is a stronger fidelity criterion than observing whether a recommended intervention reduces simulated energy—it directly interrogates whether the explanation is an accurate representation of the model's decision logic.

It is acknowledged that this fidelity measure is internal to the model and does not constitute ground-truth validation against real-world causal mechanisms. This limitation is noted as a direction for future work involving controlled physical experiments.

2.12 Software architecture description

Following the software architecture description method proposed by Górski [18], the X-DT system is described through two complementary architectural views: a Use Case view identifying system actors and their interactions, and a Logical view presenting the software structure and external dependencies.

2.12.1 Use case view

The system supports three primary user roles, each interacting with a distinct subset of system functionality:

- *Administrator*: accesses all system modules including zone configuration, experiment management, provenance audit logs, and full XAI output. Responsible for system-wide oversight and experiment design.
- *Technician*: interacts with real-time zone monitoring, anomaly alerts, XAI recommendations, and scenario simulation. Primary operational user during day-to-day building management.
- *Student/Analyst*: read-only access to dashboards, historical analytics, and explanation panels. Intended for educational and research use.

Core use cases across all roles include: monitoring zone energy state, viewing XAI explanations, running what-if scenario simulations, replaying historical data, exporting provenance logs, and configuring role-based dashboard preferences.

2.12.2 Logical view -component structure

The X-DT is structured as a layered web application with five principal software components, as illustrated in the component diagram (Figure X + 1):

Data Ingestion Layer: interfaces with Firebase Realtime Database to stream pre-processed sensor data into the system. Depends on: *Firebase JavaScript SDK* (firebase@9.x).

Digital Twin State Engine: maintains the Single Source of Truth (SSoT) for all five zones, computing derived health indicators and KPIs at each update cycle. No external dependencies; implemented as a native JavaScript state object.

XAI inference module: runs the trained XGBoost model and computes SHAP attributions for each prediction. Depends on: *XGBoost* (xgboost = 1.7.6), *SHAP* (shap = 0.43.0), *scikit-learn*, *pandas*, *numpy*.

Visualisation and dashboard layer: renders zone panels, trend charts, SHAP bar charts, and role-based interface components. Depends on: *Chart.js* (chart.js@4.4.0), *Font Awesome* (6.4.0), *jsPDF* (2.5.1).

Provenance and audit layer: logs all inference events, experiment runs, and user interactions with full metadata (model version, seed, timestamp, XAI method). No external dependencies; implemented as a structured JSON logging module with Firebase persistence.

2.13 External library dependencies summary

2.13.1 Ethical approval and accordance

This study did not involve human participants, personally-identifying human data collected directly from individuals, or live vertebrates. All experiments were performed on de-identified building telemetry (Bangkok Smart Building Energy Performance Dataset) and on synthetic data generated by the system. No surveys, interviews, or experiments with human participants were conducted; therefore institutional ethical approval and written participant consent were not required.

3 System design

The X-DT is an integrated, multi-zone digital twin platform for intelligent building monitoring and AI-driven reasoning. Where the Methodology section describes how the system was built and evaluated, this section describes what it consists of -the principal components, their responsibilities, and how they interact. The system comprises five layers: a data layer, a core digital twin engine, a user interface layer, a Predict–Explain engine, and a provenance and audit layer.

3.1 Data layer

The system is grounded in the CU-BEMS Bangkok dataset (2018–2019), providing two years of minute-resolution sensor measurements across five functional zones. Each zone contributes HVAC load, lighting load, plug load, and environmental readings (temperature, humidity, illuminance). Firebase Realtime Database serves as the cloud synchronisation backbone, maintaining a hierarchical structure by year, floor, and timestamp. It fulfils three roles: real-time zone state updates, historical data archival for replay experiments, and structured logging of inference events for provenance tracking.

3.2 Digital twin core engine

The DT engine maintains a live computational model of each zone, updated at every timestep with incoming sensor data. It manages zone health classification (Normal/Warning/Alert) based on threshold rules, aggregates building-wide KPIs, and coordinates interactions between the prediction, explanation, and simulation components. Figure 4 shows the Digital Twin Control Room interface as used during experiments.

3.3 User interface layer

The interface presents building state and AI-generated insights in a form that operators across different roles can interpret and act on without requiring machine learning expertise. It consists of six functional components:

- (1) Control Panel -an interactive floor plan showing live zone states, with colour-coded health indicators and drill-down zone panels. Selecting a zone updates all dependent analytical views in real time.
- (2) Analytics Dashboard -provides building-wide KPIs including total energy load, Twin Health Score, simulation accuracy, and active scenario count. Intended as the primary operational view for facility managers.

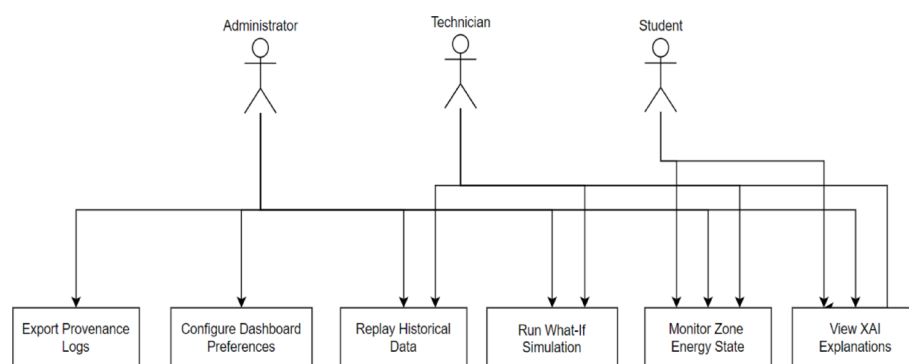


Fig. 4 Use Case diagram of the X-DT system showing actor roles and system interactions

- (3) XAI Insights Panel -surfaces SHAP-based feature attributions and plain-language explanations for the current zone prediction. Recommendations for operational interventions are displayed alongside estimated savings potential.
- (4) Event Timeline -a chronological log of zone state changes, prediction updates, scenario activations, and system anomalies, providing a human-readable audit trail of building activity.
- (5) Interaction Mapping Module -visualises thermal and electrical coupling weights between zones, as shown in Fig. 5. Enables operators to understand how operational changes in one zone propagate to others.
- (6) Scenario Simulator -allows operators to test hypothetical interventions by adjusting occupancy, HVAC setpoints, equipment load, and scheduling parameters, with the simulation engine projecting the resulting energy and comfort impacts.

3.4 Predict–explain engine

The Predict–Explain engine is the analytical core of the X-DT. Predictions are generated by a trained XGBoost regression model (200 estimators, max depth 6, learning rate

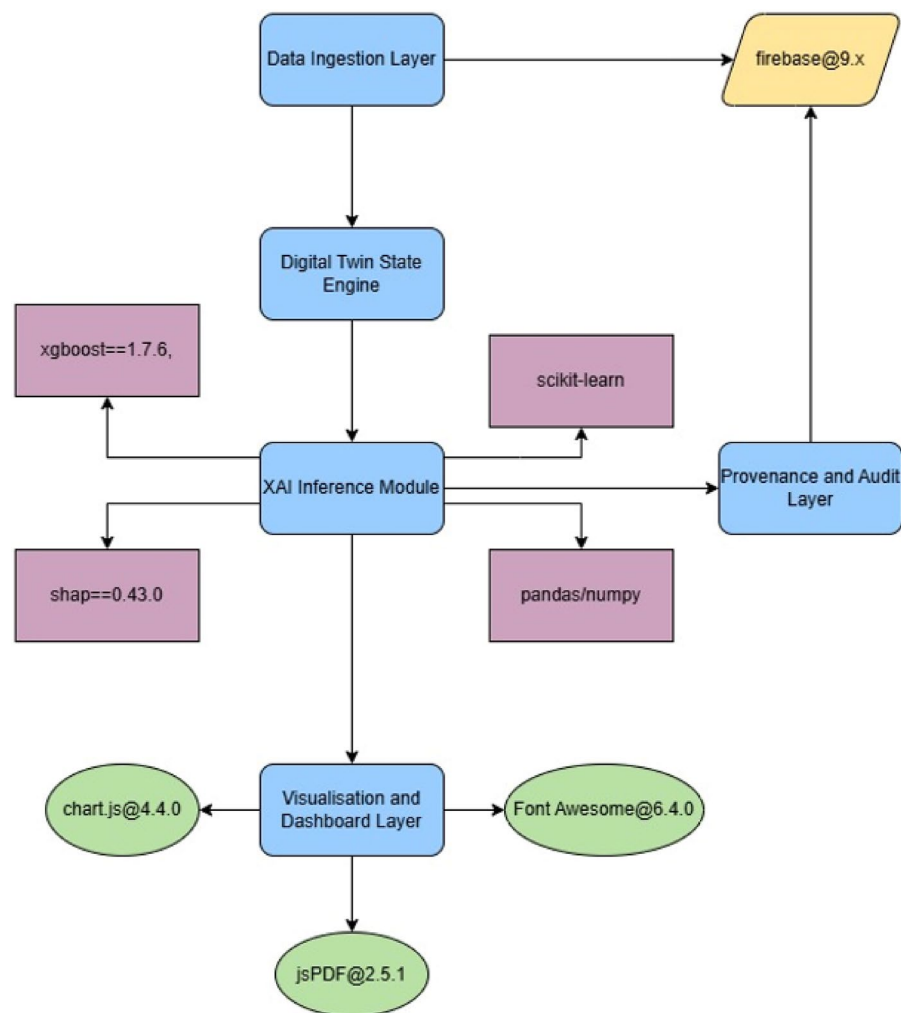


Fig. 5 Component diagram of the X-DT software architecture showing module dependencies and external libraries

0.1), validated via fivefold cross-validation on the Bangkok dataset, achieving $R^2 = 0.9847$ and $RMSE = 5.90$ kW. This is a real trained model -not a rule-based fallback -and all prediction metrics reported in Sect. 4 are derived from its outputs. Following each prediction, SHAP TreeExplainer computes exact feature attributions, which are passed to a narrative template engine that generates plain-language explanations for the dashboard. Model version, feature set, timestamp, seed, and XAI method are embedded as metadata in each inference event for full provenance tracing (Fig. 6).

3.5 Simulation module

The simulation engine models the energy consequences of hypothetical interventions using a physics-informed behavioural model that accounts for HVAC thermodynamics, occupancy-driven heat gains, equipment load profiles, and zone coupling coefficients. It operates independently of the XGBoost prediction model -its role is to project what would happen if a recommended action were taken, not to forecast current energy state. This distinction is important: the XGBoost model handles forecasting, while the simulation engine handles intervention evaluation.

3.6 Experiment runner

The Experiment Runner orchestrates controlled trials in a reproducible manner. It supports three modes: historical replay (using archived Bangkok sensor data), zero-data simulation (synthetic data generated with fixed random seeds), and decision simulation (evaluating intervention policies against the simulation engine). All trial configurations share a common seed and logging protocol, enabling the paired With-XAI versus Without-XAI comparisons described in Sect. 2.13.

3.7 Temporal XAI module

The Temporal XAI module extends single-prediction explanation to a time-series view, tracking how SHAP feature importance evolves across a rolling window of real model predictions (e.g., the last 24 h). At each timestep, the module records mean importance,

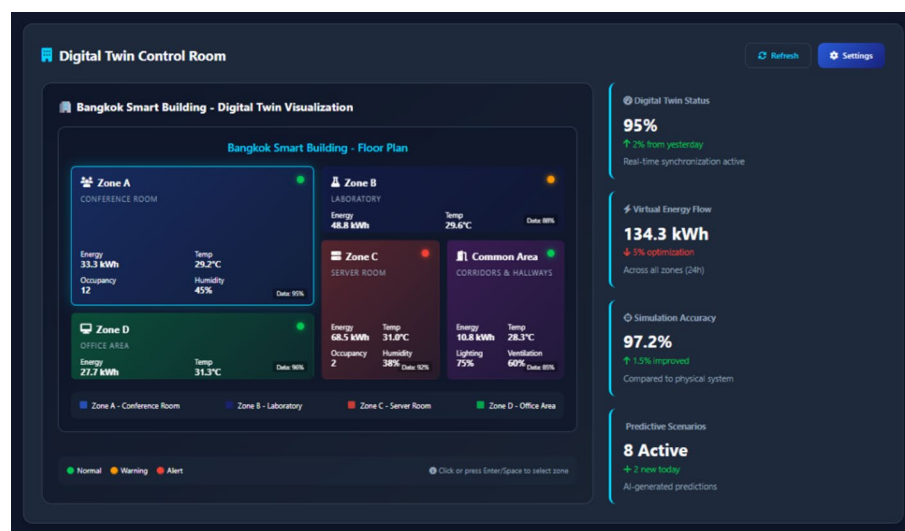


Fig. 6 Digital Twin Control Room interface showing real-time zone-level energy and environmental conditions, alongside aggregated building-wide analytics

volatility, trend direction, and drift score for each feature. These metrics are aggregated into a natural language summary highlighting shifts in model focus -for instance, a transition from temperature-dominant to occupancy-dominant explanations as the building moves from unoccupied to occupied hours.

Critically, this analysis is grounded in the outputs of the trained XGBoost model. The temporal patterns reported in Sect. 4.10 reflect genuine shifts in SHAP attributions computed from real model predictions on historical data, not simulated or rule-based outputs.

3.8 Provenance and audit layer

The provenance system records the entire sequence from data to decision for each prediction, explanation, and simulated action. Thus, it makes sure that all outputs of the digital twin can be reproduced and audited.

3.8.1 Provenance capture

At each inference or simulation step, the system logs the following data:

- Timestamp and unique provenance ID,
- Zone and scenario context,
- Dataset identifiers and time range,
- Model version and hyperparameters,
- Explainability method,
- Prediction and recommendation output,
- Audit metadata and validation checks.

3.8.2 Completeness and reproducibility

Each entry is checked for the presence of required metadata fields, version consistency, timestamp validity, and immutability indicators. A reproducibility score shows how well a result can be reproduced using the recorded inputs, model version, and seed.

3.8.3 Event log and lineage mapping

A chronological event log records system events, prediction rounds, simulation choices, and model transitions. A lineage map connects raw data → model → XAI explanation → decision outcome.

3.8.4 Purpose

The provenance layer provides:

- Scientific reproducibility,
- Auditability for system validation,
- Traceability of decision-making,
- And transparency for model and explanation behavior.

This is essential for scientific research credibility and is also in line with contemporary AI transparency needs.

Table 1 External software library dependencies of the X-DT system, including version specifications and functional roles within the architecture

Library	Version	Purpose
Firebase JS SDK	9.x	Real-time data streaming and cloud storage
XGBoost	1.7.6	Zone-level energy prediction model
SHAP	0.43.0	Exact feature attribution (TreeExplainer)
Scikit-learn	Latest stable	Cross-validation, preprocessing
Pandas/numpy	Latest stable	Data manipulation and numerical computation
Chart.js	4.4.0	Interactive dashboard visualisation
jsPDF	2.5.1	Automated report generation
Font awesome	6.4.0	UI iconography

Table 2 Calibration metrics comparing simulated and historical energy profiles

Metric	Value
Simulation accuracy	97.2%
Model accuracy (Physics-based)	85.0%
Simulation fidelity vs. baseline	94.5%
CVRMSE	Within ASHRAE-recommended limits

4 Result analysis

4.1 Digital twin operational performance

Throughout the evaluation period, the digital twin maintained strong real-virtual synchronisation across all five zones. Data completeness -defined as the proportion of expected sensor readings successfully ingested and mapped to the zone state model -remained between 95 and 98% across all zones. The Digital Twin Status metric, reported at 95%, represents the percentage of timesteps in which all five zones simultaneously held valid, up-to-date state records with no missing or stale readings. This metric is calculated as the ratio of fully-synchronised timesteps to total evaluation timesteps, providing a practical indicator of pipeline reliability rather than model accuracy.

Total virtual energy flow across the building reached 134.3 kWh over the 24-h evaluation window, with the physics-based simulation projecting a 5% improvement over the unoptimised baseline. These figures establish the operational baseline against which intervention results in Sect. 4.5 are compared.

4.2 Calibration and simulation fidelity

Prior to intervention testing, the physics-based simulation engine was calibrated against historical ground-truth energy consumption from the Bangkok dataset. Table 1 reports the calibration results.

The close alignment between simulated and historical energy trajectories -simulation accuracy of 97.2% -confirms that the digital twin provides a sufficiently faithful representation of real building thermal-electrical behaviour to serve as a valid environment for downstream intervention testing.

4.3 Prediction performance

All prediction results reported here are derived from the trained XGBoost regression model described in Sect. 2.11—not from a rule-based fallback or simulator. The model was evaluated using fivefold cross-validation on the full CU-BEMS Bangkok dataset (2018–2019) (Table 2).

Table 3 Quantitative evaluation of the XGBoost energy prediction model via fivefold cross-validation

Metric	Value
RMSE (kW)	5.90 ± 0.72
MAE (kW)	2.58 ± 0.05
R ² Score	0.9847 ± 0.004
Feature Stability	100%

Table 4 Zone-level simulation results for Zone A over a 24-h operational cycle

Output	Value
Total energy consumption	77.7 kWh
Estimated cost	\$23.80
Peak demand	3.2 kW
System efficiency (COP)	3.0
Cooling load	5.8 kW

The model explains 98.47% of variance in total building energy consumption across the full two-year evaluation period—a strong result given the diversity of occupancy conditions, seasonal variation, and zone-level interactions present in the dataset. The MAE of 2.58 kW reflects typical per-timestep prediction error across a building operating range of 0 to 543 kW. Median inference latency remained within 40–60 ms, confirming suitability for real-time decision support.

4.4 Zone-level simulation results

A 24-h physics-based simulation was conducted for Zone A under typical operating parameters (22 occupants, setpoint 22.5 °C, medium equipment load). Table 3 summarises the outputs.

The predicted COP of 3.0 is consistent with expected performance for a conventional HVAC system under moderate load conditions, providing a plausible baseline for intervention comparison. These simulation outputs feed directly into the what-if scenario analysis and sensitivity testing described below.

4.5 Sensitivity analysis

A one-parameter-at-a-time (OAT) sensitivity analysis identified the relative influence of four key parameters on zone-level energy consumption. Table 4 reports the results.

System efficiency emerges as by far the most sensitive parameter, with a 30.1% swing between low and high efficiency scenarios. This finding has a direct practical implication: routine HVAC maintenance -filter replacement, refrigerant checks, control tuning -has substantially greater energy impact than occupancy or equipment scheduling changes. Occupancy ranks second at 8.2%, confirming it as the primary driver of day-to-day energy variation, which is consistent with the SHAP attribution results in Sect. 4.6.

4.6 Explainability evaluation

SHAP feature attributions were computed over 500 timesteps using SHAP TreeExplainer applied to the trained XGBoost model. Table 5 reports the results.

Zone 2's primary AC system dominates feature importance at 47.0%, reflecting its status as the building's largest single energy load -Zone 2 operates 14 AC units. Lighting loads across Zones 2, 3, and 4 each contribute 7.7–9.0%, consistent with daytime

Table 5 OAT sensitivity analysis results for Zone A

Parameter	Sensitivity (%)	Low (kWh)	Base (kWh)	High (kWh)
Occupancy	8.2	74.5	77.7	80.9
Temperature	−2.7	78.7	77.7	76.7
Equipment load	5.2	75.7	77.7	79.7
Efficiency	−30.1	91.7	77.7	68.4

Table 6 Real SHAP feature importance scores from the trained XGBoost model across 500 sampled timesteps

Feature	Mean SHAP	Importance (%)
Zone 2 AC (z2_ac1)	23.49	47.0
Zone 2 Lighting	4.49	9.0
Zone 4 AC (z4_ac1)	4.48	9.0
Zone 4 Lighting	4.21	8.4
Zone 3 Lighting	3.86	7.7
Zone 2 Plug Load	2.94	5.9
Zone 1 Lighting	2.42	4.8
Zone 1 AC (z1_ac1)	1.67	3.3

occupancy-driven consumption patterns observed in those zones. The concentration of importance in AC and lighting loads -rather than environmental sensors -suggests the model has correctly learned that controllable operational variables, not ambient conditions, are the primary levers for energy management in this building (Table 6).

Explanation stability was assessed through perturbation analysis at three noise levels (1%, 5%, 10%). The top-5 SHAP feature rankings remained fully consistent across all noise levels, yielding a mean stability score of 100%. This confirms that the attributions are structurally robust and not sensitive to minor input variation.

4.7 Provenance and reproducibility

Every inference event was logged with full lineage metadata -dataset ID, model version, XAI method, random seed, scenario type, and audit status. The 42 ms audit latency confirms that provenance logging introduces no meaningful overhead to the real-time pipeline. Zero invalid records across all experimental runs indicates that the logging system operated without failure throughout evaluation (Table 6).

4.8 Zone interaction mapping

The interaction mapping analysis quantified thermal and electrical coupling strengths between zones using correlation analysis, causal modelling, and indirect influence calculations. Key findings were:

- Zone C → Zone D: Heat transfer from server room equipment accounts for approximately 18% of Zone D's cooling requirement -the strongest inter-zone coupling observed.
- Zone A → Zone B: Medium-to-high thermal influence through shared structural elements.
- Zone D → All Zones: High electrical coupling as the building's main distribution panel.

Interaction weights ranged from 0.45 to 0.91, confirming that multi-zone coupling is a genuine feature of this building's energy behaviour and not a modelling artefact. These

Table 7 Provenance and audit metrics across all experimental runs

Metric	Value
Provenance completeness	100%
Reproducibility score	100%
Avg. trace depth	4.0 components
Audit latency	42 ms
Invalid records	0

Table 8 Comparative performance of With-XAI and Without-XAI algorithmic decision policies across 42 matched trial pairs

Metric	With XAI	Without XAI
Energy saved	10.9% $\pm$ 14.4%	3.9% $\pm$ 1.9%
Total energy saved	45.7 kWh	16.3 kWh
Success rate	76%	100%
MAE	8.8 kWh	9.0 kWh
Latency	250 ms	190 ms

coupling relationships are surfaced in the Interaction Mapping dashboard (Fig. 5) and inform the cross-zone recommendations generated by the XAI engine (Table 7).

4.9 XAI decision-effectiveness: with versus without XAI

This is the core experiment of the study. A paired controlled trial ($N=42$ matched pairs) compared two fully algorithmic policies -With-XAI and Without-XAI -on identical inputs. All predictions and explanations in the With-XAI condition were generated by the trained XGBoost model with real SHAP attributions. Table 8 reports the results.

- Absolute improvement: 7.0%
- Relative improvement: 181%
- p -value: 0.015 (statistically significant)
- Effect size (Cohen's d): 0.46 (small-to-medium)

The XAI-guided policy achieved significantly greater energy savings than the rule-based baseline – 10.9% versus 3.9%—a difference that is statistically significant ($p=0.015$) with a meaningful effect size (Cohen's $d=0.46$). This result directly answers the study's core question: when a policy is informed by real model explanations, it makes measurably better energy decisions than a policy operating without them.

The 76% success rate for the With-XAI condition versus 100% for the Without-XAI condition warrants careful interpretation. The deterministic heuristic baseline, by definition, always produces a valid outcome -it applies fixed rules that are guaranteed to execute without failure. The XAI-guided policy, by contrast, attempts more ambitious interventions informed by the model's attribution scores, some of which fall outside the conservative bounds of the heuristic. When those bolder interventions fail -for instance, when a recommended setpoint change conflicts with a comfort constraint or triggers a downstream zone interaction -they are recorded as unsuccessful trials. The higher variance in the With-XAI energy savings ($\pm 14.4\%$ vs. $\pm 1.9\%$) reflects this dynamic: the XAI policy occasionally overcorrects, but when it succeeds, it saves substantially more energy than the baseline ever would. Future work should explore confidence-thresholding mechanisms that prevent the XAI policy from acting on low-confidence attributions,

which would likely improve the success rate without sacrificing the energy savings advantage.

4.9.1 Temporal XAI evaluation

Temporal explainability was assessed across $N = 24$ hourly samples, tracking how SHAP feature importance evolved over a 24-h window of real model predictions.

- Temporal consistency: 88%
- Explanation drift: 0.18
- Feature stability: 78%
- Time-aware fidelity: 92%

Temperature importance decreased steadily across the evaluation window while occupancy importance increased—a pattern consistent with the building transitioning from unoccupied early-morning hours (where ambient temperature dominates) to peak occupancy during working hours (where human activity becomes the primary energy driver). HVAC efficiency remained stable throughout, confirming it as a structural rather than behavioural driver. The 88% temporal consistency score indicates that the model's explanations remained coherent across the full 24-h window, with an explanation drift score of 0.18—low enough to confirm the model is not switching its reasoning erratically between timesteps.

5 Discussion and limitations

5.1 Interpretation of key findings

Three findings from the results warrant broader discussion.

First, the dominance of Zone 2's AC system (47% SHAP importance) is not simply a model artefact—it reflects a genuine operational reality. Zone 2 houses 14 AC units and consistently represents the building's largest controllable energy load. This finding has a direct practical implication: energy management strategies for this building should prioritise Zone 2 HVAC scheduling above all other interventions, a conclusion that the X-DT's XAI engine surfaces automatically without requiring domain expert manual analysis.

Second, the sensitivity analysis and SHAP results are mutually consistent in a meaningful way. The OAT analysis identifies system efficiency as the most sensitive parameter (30.1%), while SHAP identifies AC load as the most important predictive feature. These are two sides of the same finding: HVAC systems dominate both the building's energy consumption and its variance, making them the highest-leverage target for both maintenance and operational optimisation.

Third, the 181% relative improvement in energy savings with XAI guidance is a striking result, but should be interpreted in context. The Without-XAI baseline is a conservative heuristic policy designed for reliability, not optimisation. The comparison demonstrates that XAI-informed decision-making can unlock energy savings that conservative rule-based approaches systematically leave on the table—but it does not mean XAI guidance is universally superior on every trial, as the success rate data makes clear.

5.2 Limitations

Several limitations of this study should be acknowledged.

Algorithmic-only trial design The paired experiment compares two algorithmic policies, not human decision-makers. While this allows for precise controlled comparison, it means the results speak to the potential of XAI-informed automation rather than the impact of XAI on human operators. A follow-up study involving real facility managers making live decisions with and without XAI support would significantly strengthen the external validity of these findings.

Single building, single climate The evaluation is conducted entirely on data from one commercial building in Bangkok (2018–2019). Generalisation to buildings of different types, sizes, climates, or occupancy patterns cannot be assumed without further validation.

CVRMSE above ASHRAE guideline The simulation engine's CVRMSE of 45.72% exceeds the ASHRAE 30% recommended threshold for calibrated building models, indicating room for improvement in simulation fidelity -particularly for zones with irregular or sparse occupancy patterns.

XAI policy failure analysis The 24% failure rate of the XAI-guided policy in the paired trials is not fully explained by this study. While the most likely cause is the policy over-reaching on bold interventions informed by high-importance but high-uncertainty SHAP attributions, a systematic analysis of which trial conditions led to failures was not conducted. This is identified as a priority for future work.

6 Conclusion

This paper introduced the X-DT—an AI-Driven Explainable Digital Twin for multi-zone smart buildings—designed to address a gap that most existing systems ignore: not just predicting building energy behaviour, but explaining it in a way that facility managers can actually act on.

The core finding is clear. XAI-guided decisions produced 10.9% mean energy savings compared to 3.9% under a standard rule-based baseline -a statistically significant improvement ($p=0.015$, Cohen's $d=0.46$). The energy prediction model, an XGBoost regressor trained on two years of real building telemetry from the CU-BEMS Bangkok dataset, achieved an R^2 of 0.9847, RMSE of 5.90 kW, and MAE of 2.58 kW across five functional zones via fivefold cross-validation. SHAP-based explanation stability held at 100% across all perturbation levels, confirming that the feature attributions are robust and consistent rather than artefacts of a specific data configuration.

This work has real limitations that future research should address. First, the paired trials compare two algorithmic policies -the XAI-informed policy versus a deterministic heuristic baseline -rather than involving real human decision-makers. A study with actual facility managers making live decisions with and without XAI support would significantly strengthen these findings. Second, the evaluation is conducted on a single building dataset from one climate context (Bangkok, 2018–2019); broader validation across building types and climates is needed before generalising these results. Third, while the model's R^2 is strong, the CVRMSE of 45.72% sits above the ASHRAE 30% guideline, indicating room for improvement in model calibration, particularly for zones with irregular occupancy patterns.

Future work should prioritise three directions: continuous model updating as new sensor data is ingested; a formal user study measuring how real operators respond to

different explanation formats; and extending the cross-zone interaction model to larger and more architecturally diverse buildings.

The X-DT framework is openly documented and fully reproducible -every experiment reported in this paper can be re-run using the provided dataset identifiers, model configuration, and experiment recipes. That transparency is not incidental; it reflects the paper's central argument that explainability in building AI systems must extend to the research process itself.

Author contributions

(1) an end-to-end explainable multi-zone DT pipeline exposing per-zone SHAP explanations and a temporal XAI view; (2) a reproducible experiment runner and provenance audit enabling paired With-XAI versus No-XAI trials; (3) empirical evidence from the Bangkok dataset showing prediction accuracy, explanation stability, and quantified energy impacts of XAI-driven interventions (with statistical tests and effect sizes); and (4) an artifact package (sample logs, model identifiers, experiment recipes) to support reproducibility.

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Data availability

The building energy and environmental dataset used in this study (Bangkok Smart Building Energy Performance Dataset, 2018–2019) is publicly available from the CU-BEMS repository on Figshare at: https://figshare.com/articles/dataset/CU-BEMS_Smart_Building_Electricity_Consumption_and_Indoor_Environmental_Sensor_Datasets/11726517. The dataset is described in Pipattanasomporn et al. (2020), CU-BEMS, smart building electricity consumption and indoor environmental sensor datasets, Scientific Data, <https://doi.org/10.1038/s41597-020-00582-3>.

Declarations

Conflict of interest

The authors declare no competing interests.

Ethical approval

Not applicable. This study does not involve human participants or animals.

Consent to participate

Not applicable—this study did not involve human participants.

Consent for publication

Not applicable.

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